



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE  
APPLICATIONS AND GEO-INFORMATICS**

**WEEKLY PROGRESS REPORT (25/05/2026 – 31/05/2026)**

**WEEK 2**

**PROJECT NAME**

**ROAD CONDITION DETECTION**

**PROJECT DESCRIPTION :**

**AUTOMATIC DETECTION AND  
SEGMENTATION OF ROAD DAMAGES**

**GROUP MEMBER :**

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**GROUP ID :**

**2026G263**

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|                   |                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>25/05/2026</b> | Updated the road damage dataset with additional samples and reviewed annotations to improve class representation and model accuracy.                 |
| <b>26/05/2026</b> | Retrained the YOLOv11 segmentation model using the updated dataset and monitored training metrics and validation results                             |
| <b>27/05/2026</b> | Tested the trained model on road videos and analyzed detection performance for potholes, cracks, ravelling, and other classes.                       |
| <b>28/05/2026</b> | <b>Holiday</b>                                                                                                                                       |
| <b>29/05/2026</b> | Investigated model misclassifications, particularly confusion between trees and ravelling, and studied methods to improve detection accuracy.        |
| <b>30/05/2026</b> | Optimized the inference pipeline in Google Colab, resolved model loading and video processing issues, and evaluated results on multiple test videos. |
| <b>31/05/2026</b> | <b>Holiday (Sunday)</b>                                                                                                                              |

|                       |                                                                                                                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Next Week Plan</b> | Improve road damage detection accuracy by applying road-region association rules, removing bounding-box visualization, reducing false positives, and optimizing model predictions. |
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## REFERENCE:

- RDD2022 Road Damage Dataset (Arya et al., 2024).
- YOLO-based Road Damage Detection Studies (Ma et al., 2025; Wang et al., 2025).
- Ultralytics YOLO Documentation.
- SEA-YOLOv8 Road Surface Damage Detection Framework (Zhao et al., 2025).

## SCREEN SHOTS:

